# Practice workbook · Network topologies: bus, ring, star and mesh

Recognising a topology from a diagram or description and stating/explaining its benefits, drawbacks and fault tolerance.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-05

# 1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

**1** There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

**(2)**

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

**(1)**

.....

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P 7 5 7 3 6 A 0 2 2 8

# 1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

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(c) **Figure 1** shows a network topology.

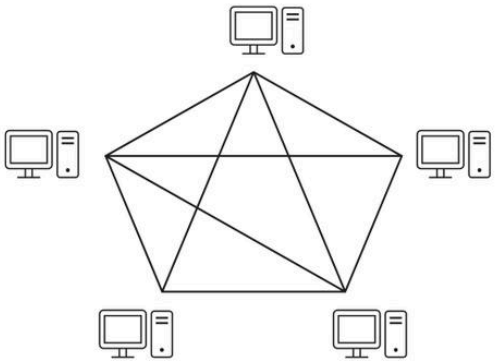


**Figure 1**

(i) State the name of this network topology.

**(1)**

(ii) State **one** drawback of using this network topology.

**(1)**

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

**(2)**

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# 1. Question 1 · part c.iii · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part c.iii only.

(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

**(Total for Question 1 = 10 marks)**

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Working and answer

## 2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

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**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

.....

.....

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

.....

.....

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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## 2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ☒ | ☒ | ☒ | ☒ |
| All devices are directly connected to all others. | ☒ | ☒ | ☒ | ☒ |
| Each node has a physical attachment to a routing device. | ☒ | ☒ | ☒ | ☒ |
| There are no data collisions, because packets travel in the same direction. | ☒ | ☒ | ☒ | ☒ |

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## 2. Question 3 · part d · 4 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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## 2. Question 3 · part d · 4 marks · Working and answer

Working and answer

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## 3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

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2

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## 3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

**(2)**

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

**(2)**

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## 3. Question 1 · part e.i · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.i only.

(e) Network layouts can be based upon a number of different network topologies.

**Figure 1** shows a network topology.

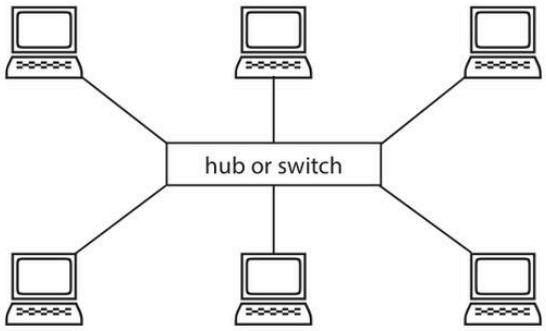



**Figure 1**

(i) State the name of this network topology. (1)

(ii) State **one** benefit of using this network topology. (1)

(iii) State **one** drawback of using this network topology. (1)

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## 3. Question 1 · part e.i · 1 mark · Working and answer

Working and answer

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## 4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

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2

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## 4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

**(2)**

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

**(2)**

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# 4. Question 1 · part e.ii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.ii only.

(e) Network layouts can be based upon a number of different network topologies.

**Figure 1** shows a network topology.

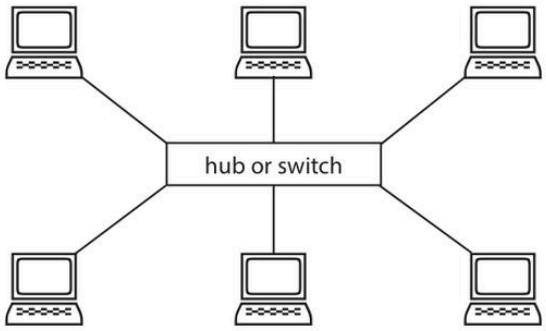



**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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## 4. Question 1 · part e.ii · 1 mark · Working and answer

Working and answer

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## 5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

**1** Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

**(2)**

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

**(2)**

1 .....

.....

2 .....

.....

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2

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## 5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

**(2)**

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

**(2)**

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## 5. Question 1 · part e.iii · 1 mark

2020 NOV · Paper 01

Further practice · Attempt part e.iii only.

(e) Network layouts can be based upon a number of different network topologies.

**Figure 1** shows a network topology.

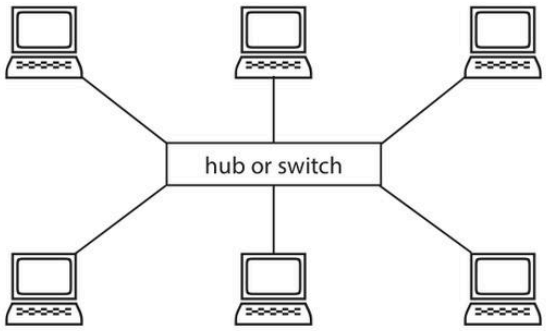



**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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DO NOT WRITE IN THIS AREA



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## 5. Question 1 · part e.iii · 1 mark

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Further practice · Attempt part e.iii only.

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(f) The network uses TCP/IP.

**Figure 2** shows the TCP/IP protocol stack.

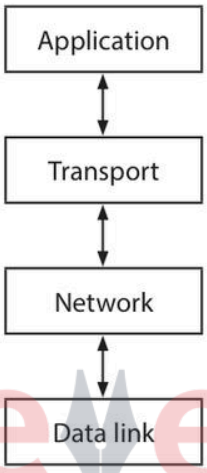



**Figure 2**

(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

**(2)**

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

**(2)**

1 .....

.....

2 .....

.....

**(Total for Question 1 = 17 marks)**

## 5. Question 1 · part e.iii · 1 mark · Working and answer

Working and answer

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# 6. Question 2 · part e.i · 1 mark

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Further practice · Attempt part e.i only.

DO NOT WRITE IN THIS AREA

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**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

(b) State the purpose of network protocols.

**(1)**

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

**(1)**

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

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P 6 6 4 8 9 A 0 5 2 0

# 6. Question 2 · part e.i · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.i only.

(e) **Figure 2** shows a network topology.

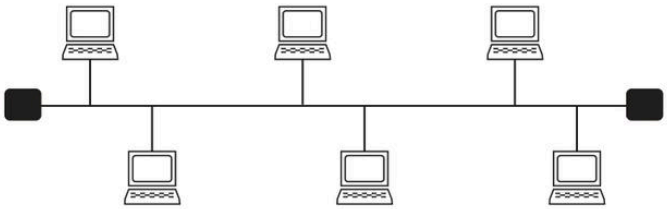


**Figure 2**

(i) State the name of this network topology.

**(1)**

.....

(ii) State **one** benefit of using this network topology.

**(1)**

.....

.....

(iii) State **one** drawback of using this network topology.

**(1)**

.....

.....

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

1 .....

.....

2 .....

.....

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DO NOT WRITE IN THIS AREA



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## 6. Question 2 · part e.i · 1 mark · Working and answer

Working and answer

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# 7. Question 2 · part e.ii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.ii only.

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DO NOT WRITE IN THIS AREA

**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

(b) State the purpose of network protocols.

**(1)**

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

**(1)**

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
| https://www.pearson.com | | | |
| 216.33.251.195 | | | |
| pearson.com | | | |

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# 7. Question 2 · part e.ii · 1 mark

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Further practice · Attempt part e.ii only.

(e) **Figure 2** shows a network topology.

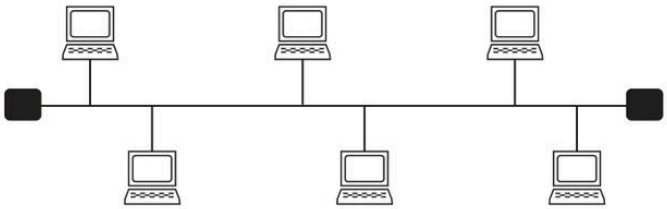


**Figure 2**

(i) State the name of this network topology.

**(1)**

(ii) State **one** benefit of using this network topology.

**(1)**

(iii) State **one** drawback of using this network topology.

**(1)**

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

1 .....

2 .....

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## 7. Question 2 · part e.ii · 1 mark · Working and answer

Working and answer

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## 8. Question 2 · part e.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.iii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

**2** Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

**(1)**

.....

.....

(b) State the purpose of network protocols.

**(1)**

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

**(1)**

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

**(1)**

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

**(3)**

| Item | Domain Name | IP Address | URL |
|---|---|---|---|
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| 216.33.251.195 | | | |
| pearson.com | | | |

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# 8. Question 2 · part e.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part e.iii only.

(e) **Figure 2** shows a network topology.

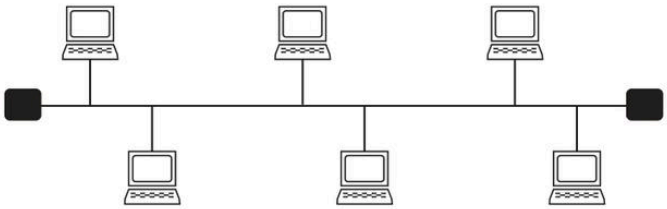


**Figure 2**

(i) State the name of this network topology.

**(1)**

.....

(ii) State **one** benefit of using this network topology.

**(1)**

.....

.....

(iii) State **one** drawback of using this network topology.

**(1)**

.....

.....

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

**(2)**

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2 .....

.....

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## 8. Question 2 · part e.iii · 1 mark · Working and answer

Working and answer

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## 9. Question 6 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

**6** Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

**Figure 3** shows a network topology.

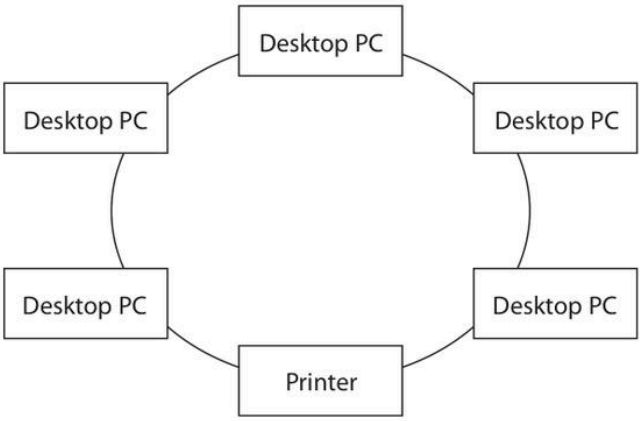



**Figure 3**

(i) Explain **one** benefit of this network topology. **(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet. **(2)**

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P 7 2 5 3 8 A 0 1 8 2 0

## 9. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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## 10. Question 6 · part a.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

**6** Santiago manages a computer network for a small business.

(a) Networks are based on a topology.

**Figure 3** shows a network topology.

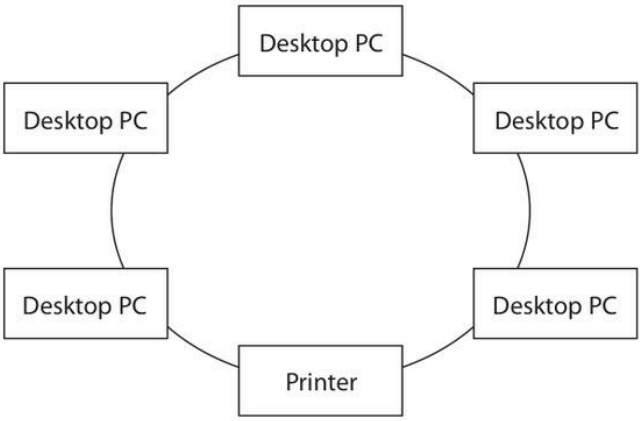



**Figure 3**

(i) Explain **one** benefit of this network topology.

**(2)**

(ii) The internet is the world's largest mesh network.

Explain **one** reason why a mesh topology is essential for the internet.

**(2)**

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DO NOT WRITE IN THIS AREA

## 10. Question 6 · part a.ii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

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(b) Santiago works on his laptop whilst travelling by train.

There is a free Wi-Fi connection on the train, but Santiago doesn't use it.

He prefers to set up a network between his smartphone and his laptop to connect to the internet.

(i) Name this type of network.

**(1)**

(ii) Explain **one** advantage for Santiago of using the network he has set up to connect to the internet, rather than the free Wi-Fi connection.

**(2)**

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(c) Santiago uses audit trails to help protect the network.

(i) State what is meant by an **audit trail**.

**(1)**

(ii) Give **one** way the data from audit trails can be used to help keep the network secure.

**(1)**

## 10. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

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## 11. Question 1 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

**1** There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

**(2)**

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

**(1)**

.....

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# 11. Question 1 · part c.i · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.i only.

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DO NOT WRITE IN THIS AREA

(c) **Figure 1** shows a network topology.

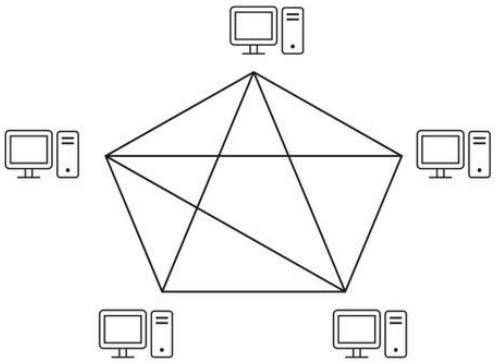


**Figure 1**

(i) State the name of this network topology.

**(1)**

(ii) State **one** drawback of using this network topology.

**(1)**

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

**(2)**

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## 11. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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## 12. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

**1** There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

**(2)**

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

**(1)**

.....

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DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

## 12. Question 1 · part c.ii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(c) **Figure 1** shows a network topology.

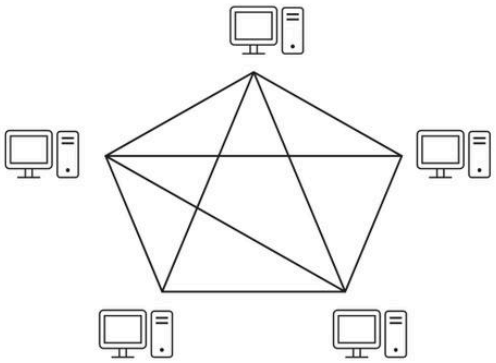


**Figure 1**

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

(2)

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## 12. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

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